Golden Scenario Coverage Matrix

*Built 2026-08-04 by Quinn (Test Architect). Every "covered" cell below was verified by reading the approved snapshot, not by reading the scenario file — a scenario that sets a field but never reaches the code path it was meant to exercise counts as **not covered**.*

35 scenarios. 405 tests green. Each scenario is one end-to-end run whose **entire** response is compared byte-for-byte with its approved snapshot.

How this matrix was built

The suite grew feature by feature, so it was deep on battery behaviour and thin everywhere else. Three checks exposed that:

1. **Which snapshot shows a non-zero value for X?** — not "which scenario sets X". This is what revealed that **Level 2 produced zero savings in all 18** original scenarios, and that **PTI assist never actually engaged** despite two scenarios named after it.
2. **Which branch of each user-facing message is reached?** — only 2 of the 4 Level 1 rejection messages had a golden scenario.
3. **Which plant shapes exist?** — 15 of 18 scenarios used the *same* plant.

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1 · Optimization levels

Path	Scenarios	Notes
L1 savings only	01–10, 12, 13, 15–18, 27–30, 32, 34	the common case
L2 savings > 0	**19, 23, 24, 25, 26, 31, 33**	needs ≥2 aux running AND a curve region where an asymmetric split wins
L3 savings > 0	11, 19, 23–26, 31, 33	
All three tiers differ	**19, 23, 24, 25, 26, 31, 33**	Advanced ≠ Pro ≠ Premium
Zero savings at every level	14	one valid combination — optimum *is* the baseline
L2 pass-through (0 or 1 aux running)	01–18, 27–30, 32	nothing to redistribute

Why L2 was invisible for so long: the aux SFOC curve falls monotonically from 228 g/kWh at 10 % to 193 at 74–80 %. More load is always better, and Level 1 already picks the fewest engines. Level 2 can only win by an **asymmetric** split — one generator at the 10 % floor, one at the 90 % ceiling — which needs a demand that a single engine cannot legally carry. Scenario 19 is built for exactly that.

2 · Operational modes

Combination	Scenarios
Transit only	01–06, 08–10, 12, 13, 15, 16, 17, 18, 19, 21, 23–26, 29, 30, 31, 33, 34, 35
Transit + DP	04, 20, 22, 32
Transit + Port	07
Transit + Anchor	**27**
Transit + Maneuvering	**28**
All five modes	11

Maneuvering is the only non-DP mode besides Transit that carries propulsion — 28 is the only scenario that exercises it alone.

3 · Plant topology

Shape	Scenarios
Shaft generator installed	01–16, 18, 27–30, 32, 34, 35
No shaft generator, successful result	**19, 23, 24, 25, 26, 33**
No shaft generator, rejected	17, 20, 21, 22
1 main engine	14
2 main engines	all others
Optimum runs ≥2 aux engines	19, 23–26, 31, 33
Optimum runs 0 aux engines	06, 11, 14
PTI capacity configured AND engaged	**32**
PTI configured, never engaged	08, 09

PTI could only be exercised in DP. The Transit ME-utilisation validation does not account for PTI, so any Transit plant needing shaft-motor assist is rejected before Level 1 sees it. Scenario 32 puts the deficit in DP mode, where the check does not apply. *(Logged as a question — see below.)*

4 · Battery

Path	Scenarios
No battery	03, 14, 17, 19–28, 31–33, 35
Budget exhausts mid-cascade	02
Budget saturated (surplus unused)	**29**
Battery in Transit	01, 02, 05, 06, 08–10, 12, 13, 15, 16, 29, 30
Battery in DP	04
Battery in Transit + Port	07
Battery assigned to a zero-hour mode	18
Battery configured with **no** relevant modes	34
Mission heavy consumer	05, 06
DP class redundancy	04
L3 residual after battery shaving	01, 05–08, 10–13, 15, 16, 29, 30
Battery shaves the **entire** variation	06

Scenario 29 proves invariant INV-2 end-to-end: at 10 000 kW the covered band (204.40) and the spinning reserve (444.75) are **identical** to the 1 260 kW case — only the unused surplus grows.

5 · Fuels and CO_2

Combination	Scenario	ME factor	AE factor
MDO / MGO	most	3.93267	3.93267
LNG / MGO	13	2.753	3.93267
HFO / HFO	**23**	3.114	3.114
Ammonia / MGO	**24**	**0.35154**	3.93267

24 is the widest gap the model can produce — an 11× difference between the two engines' factors on one vessel. It is the strongest guard against a regression that collapses back to a single constant.

6 · Level 3 variation source

Source	Scenario	Result
Explicit `hotelLoadVariationKw`	most	as entered
Lookup → Bulk Carrier	14	250 kW
Lookup → Container	**25**	1 500 kW
Lookup → LNG	**33**	1 000 kW
Unknown vessel type → default	**26**	500 kW

All three lookup entries plus the default are now covered. 14 also proves the *substring* match ("Bulk Carrier 10,000 dwt" → "Bulk Carrier").

7 · Baseline selection

Path	Scenario
Default: highest-FOC combination	03, 17, 19–28, 31–33
Default with battery: third-highest (D1)	01, 02, 04–16, 29, 34
User-pinned index	15
Pinned index out of range → default	**30**

30 produces a snapshot identical to 01's baseline block — which is the assertion: an index of 99 must be ignored, not clamped to the last row.

8 · Failure and advisory paths

Response	Scenario	What it proves
400 · ME utilisation > 100 %	17	validation, before Level 1
400 · battery PTI discharge gate	09	`RejectionTally` branch 1
400 · engines cannot carry the demand	**20**	branch 2
400 · aux above 90 % load	**21**	branch 3
400 · no configuration can cover	**22**	branch 4 (structural fallback)
400 · several errors at once (5)	**35**	the error list and its order
200 + battery capacity warning	10	
200 + operating-hours warning	14	profile of 8 935 h
200 + two advisory warnings	**34**	battery without modes · DP redundancy without DP

All four `Level1RejectionTally.ExplainFor` branches now have end-to-end coverage. The wording of each is additionally pinned by `Level1RejectionDiagnosticsTests`.

What is still NOT covered — deliberately

Gap	Why it is left
Every individual validation message	25+ of them; `ValidationServiceTests` pins each. The golden suite covers the *response shape*, not every string.
Sail changing the chosen combination	12 covers the sail path; a plant where sail flips the optimum would be contrived.
A vessel with a mixed-unit category	Backend logs a warning; no client-visible effect.
Leap-year hours (8 784)	The model is annual; 8 760 is the stated constant.

Open questions this exercise raised

1. **The 10 % / 90 % split.** In scenarios 19/23–26/31/33 Level 2's answer is one generator at the minimum and one at the ceiling. A crew would more likely run a single generator at 100 % — which the 90 % ceiling forbids. **Does the reference workbook agree with 10/90?** If not, the floor or the ceiling may be mis-set.
2. **PTI can never engage in Transit**, because the ME-utilisation validation ignores PTI capacity. Intended, or an oversight? Today it means the shaft-motor feature is DP-only in practice.
3. **Scenario 14 remains 175 h over a year.** It now carries the advisory warning rather than being corrected, so the scenario also documents what an over-long profile looks like.

Approval status

Scenarios 01–18 were verified against the reference workbook during the original QA passes.

Scenarios 19–35 were generated from the current code. They are *characterisation* snapshots: they prove the behaviour does not change silently, and each one was checked to actually reach the path it targets. Where an expected figure could be derived by hand it is shown in the matching card under **Hand-check**. Figures that require the workbook are marked **pending reference verification** and should be confirmed before these are treated as correctness proofs rather than change detectors.
